# MARSELINUS ALLEN NUGRAHA

**B.Eng. in Electrical Engineering • Firmware, IoT & Agentic AI Developer**


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**GitHub:** github.com/Marsel204

**Web Portfolio:** marsel204.github.io/cv-landing-page




## PROFESSIONAL SUMMARY


Undergraduate Electrical Engineering student at Universitas Pembangunan Nasional "Veteran" Jakarta (Cumulative GPA: **3.82/4.00**) specializing in *embedded firmware development* (ESP32/ESP8266, C/C++), *Internet of Things* architectures (MQTT, Modbus RTU, Blynk), industrial electrical control panel design (AutoCAD Electrical, PLC Ladder Logic), and *Agentic AI & Edge Computer Vision* engineering (LangGraph, RAG, YOLOv11, NVIDIA Jetson). Proven technical leadership with over 2 years of active service as Chairman of the IoT Student Study Group (KSM IoT FT UPNVJ) and Certified Laboratory Assistant across 4 core engineering laboratory courses, committed to delivering reliable, production-grade hardware-software automation systems.


## EDUCATION

**Universitas Pembangunan Nasional "Veteran" Jakarta** — Depok, West Java, Indonesia — Aug 2023 – Jan 2027 (Expected)

*Bachelor of Engineering (B.Eng.) in Electrical Engineering • Control, Embedded IoT & Intelligent Systems • GPA: 3.82 / 4.00*

- **Undergraduate Thesis / Research Focus:** *Design and Implementation of an Adaptive Traffic Signal Controller (ATSC) Based on YOLOv11 Multi-Object Detection and Sugeno ANFIS Traffic Controller on NVIDIA Jetson Orin Nano Edge Runtime.*
- **Relevant Coursework:** Microprocessor & Microcontroller Systems, Digital Systems, Electric Circuits, Basic Electronics, Automatic Control Systems, and Computer Programming.

**SMAN 12 Tangerang** — Tangerang City, Banten, Indonesia — Jul 2020 – May 2023

*High School Diploma in Mathematics and Natural Sciences (MIPA) • Graduated*

## ORGANIZATIONAL & LEADERSHIP EXPERIENCE

**IoT Student Study Group (KSM IoT) – Faculty of Engineering, UPN "Veteran" Jakarta** — Jan 2026 – Present

*Chairman (Ketua Umum)*

- Direct strategic management across 3 departments and 9 technical and non-technical divisions, ensuring end-to-end execution of annual engineering programs.
- Lead student research teams in national technology and robotics competitions while coordinating cross-divisional IoT hardware-software prototyping initiatives.

**IoT Student Study Group (KSM IoT) – Faculty of Engineering, UPN "Veteran" Jakarta** — Jan 2025 – Dec 2025

*Staff of Firmware Division*

- Developed embedded firmware on ESP32 and ESP8266 microcontrollers in Arduino C/C++ featuring multi-sensor integration, serial communication, MQTT telemetry, and Blynk cloud dashboards.
- Competed in national engineering competitions including **GEMASTIK** and the **Futura Transporter Robot Competition** as firmware architect responsible for low-latency control loops.

**Electrical Engineering Student Association (HMTE) – FT UPN "Veteran" Jakarta** — Jan 2025 – Dec 2025

*Staff of Research and Development Division (Litbang)*

- Designed and executed academic competency and engineering software training programs for Electrical Engineering undergraduates.
- Served as **Project Officer for Propel KTI 2025**, **Head of Scientific Paper (KTI) Division for Electrolympic 2025**, **Head of Event Division for Propel Software 2025**, and Organizing Committee Member for the **Adjunct Professor Program 2025**.

**Musicasa – SMAN 12 Tangerang** — Jan 2020 – Dec 2022

*Vice President (Wakil Ketua)*

- Spearheaded organizational revitalization following the COVID-19 pandemic, significantly increasing program continuity and active member engagement.

## ACADEMIC LABORATORY WORK & ENGINEERING PROJECTS

**Electrical Engineering Laboratory – Faculty of Engineering, UPN "Veteran" Jakarta** Aug 2024 – Present

*Certified Laboratory Teaching Assistant (Basic Physics, Digital Systems, Electric Circuits, Basic Electronics)*

- Assisted faculty lecturers in delivering hands-on laboratory practicums across 4 core Electrical Engineering courses, mentoring students in circuit analysis, digital logic synthesis, and hardware troubleshooting.
- Calibrated laboratory measurement instruments (oscilloscopes, signal generators, multimeters), prepared experimental modules, and evaluated weekly laboratory reports and practical examinations.

**Selected Embedded IoT, Industrial Automation & Agentic AI Projects** 2025 – 2026

*Lead Developer & Hardware-Software Architect • github.com/Marsel204*

- **DeploySkripsi — ATSC Edge Runtime on NVIDIA Jetson (2026):** Engineered a real-time adaptive traffic signal control system on NVIDIA Jetson Orin Nano combining YOLOv11 multi-object tracking (TensorRT FP16), *Sugeno ANFIS* phase inference, and hardware actuation over RS-485 fieldbus.
- **Industrial Root Cause Analysis / IndustrialRCA (2026):** Built a multi-agent industrial diagnostic platform powered by **LangGraph**, RAG, and DeepSeek AI compliant with ISA-95 / ISO 14224 taxonomy and live Modbus RTU telemetry ingestion.
- **MBG Food Freshness & Touchscreen GUI / fuzzylogic-tft (2026):** Developed a food quality assessment prototype integrating Electrochemical Impedance Spectroscopy (EIS), gas/pH sensors, a dual C99/Python Fuzzy Inference Engine, and an interactive ESP32-S3 TFT touchscreen GUI (GUITION JC2432W328C).
- **PLC-Based Multi-Floor Elevator & DOL Motor Starter CAD (2025):** Programmed multi-floor elevator PLC Ladder Logic in **Siemens TIA Portal & Factory I/O 3D**, and authored industrial Direct On Line (DOL) Motor Starter power/control schematics, panel layouts, and automated Bill of Materials (BOM) in **AutoCAD Electrical**.
- **Antigravity Embedded Suite & VisionLabs Traffic Annotator (2026):** Created a 4-pillar microcontroller automation framework (KiCad schematic synthesis, headless Arduino-CLI builds, NVS sensor calibration, ESP32 crash triage) and a *Label Fusion* computer vision annotation suite (GroundingDINO + YOLOv11).

## SKILLS (HARD SKILLS & SOFT SKILLS)

### Hard Skills (Technical Competencies)

- **Embedded & IoT:** ESP32 / ESP32-S3, ESP8266, NVIDIA Jetson, Arduino C/C++, C99, MQTT, Blynk, Modbus RTU / RS-485, UART/I2C/SPI, KiCad EDA.
- **Industrial Automation & CAD:** AutoCAD Electrical, PLC (Ladder Logic), Siemens TIA Portal, Factory I/O 3D, DOL Motor Starter & Panel Wiring, ISA-95.
- **AI & Software Engineering:** Python, Agentic AI (LangGraph, RAG), YOLOv11, OpenCV, PyTorch, TensorRT/ONNX, Fuzzy Logic & ANFIS, FastAPI, Git.
- **Languages:** Indonesian (Native), English (Professional Working Proficiency).

### Soft Skills (Interpersonal & Leadership)

- **Technical Leadership:** Directing 3 departments & 9 divisions in KSM IoT FT UPNVJ and faculty-level engineering committees.
- **Project & Time Management:** Coordinating parallel research milestones, laboratory instruction, and national competition deadlines.
- **Cross-Functional Teamwork:** Collaborating across hardware design, embedded firmware, and AI software engineering teams.
- **Analytical Problem Solving & Communication:** Hardware-in-the-loop debugging and technical instruction in university laboratories.

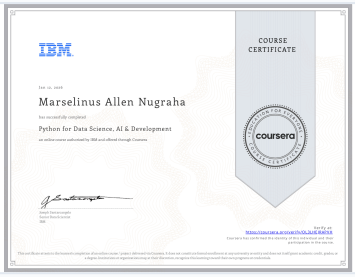
## CERTIFICATIONS, COURSES & ACHIEVEMENTS

- **Python for Data Science, AI & Development** — IBM & Coursera (Verified: Jan 2026 • Credential ID: `0L3LHEIRAPVH`).
- **Code Generation and Optimization Using IBM Granite** — IBM SkillsBuild (Verified via Credly: Oct 2025).
- **AutoCAD Electrical Software Training Program (15 Practical Hours)** — HMTE UPNVJ (Jul 2025 • Certificate No: `001/SRT-PES/PROPELSOFTWARE/UN61/HMTE/VIII/2025`).
- **Certified Laboratory Teaching Assistant (4 Practicum Courses)** — Dean's Decree, Faculty of Engineering UPNVJ: *Basic Physics* & *Digital Systems* (Even Semester 2024/2025 • `SK/4526/UN61/HK.03.01/2025`), *Electric Circuits* (Odd Semester 2025/2026 • `SGAS/159/UN61/2025/FT`), and *Basic Electronics* (Odd Semester 2025/2026 • `SGAS/160/UN61/2025/FT`).
- **Organizing Committee Member, Adjunct Professor Program** — Electrical Engineering Department, UPNVJ (Sep 2025 • Certificate No: `SERTF/546/UN61/2025/FT`).
- **National Engineering Competitions:** Firmware Division Contributor in **GEMASTIK** and the **Futura Transporter Robot Competition** (2025).

## APPENDIX 1 — VERIFIED CERTIFICATES & LABORATORY ASSISTANT DECREES

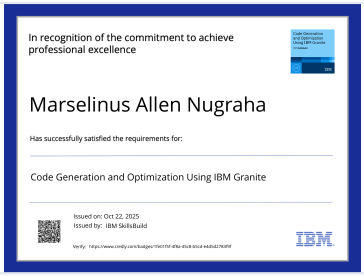
### 1. Python for Data Science, AI & Development (IBM & Coursera)

Issued: Jan 11, 2026 • coursera.org/verify/0L3LHEIRAPVH




### 2. Code Generation & Optimization Using IBM Granite

Issued: Oct 22, 2025 • IBM SkillsBuild (Verified Credly)




### 3. AutoCAD Electrical Software Training Program (15 Hours)

No: 001/SRT-PES/PROPELSOFTWARE/UN61/HMTE/VIII/2025




### 4. Lab Assistant: Basic Physics — FT UPNVJ

Even Semester A.Y. 2024/2025 • SK/4526/UN61/HK.03.01/2025




### 5. Lab Assistant: Digital Systems — FT UPNVJ

Even Semester A.Y. 2024/2025 • SK/4526/UN61/HK.03.01/2025




### 6. Lab Assistant: Electric Circuits — FT UPNVJ

Odd Semester A.Y. 2025/2026 • SGAS/159/UN61/2025/FT




### 7. Lab Assistant: Basic Electronics — FT UPNVJ

Odd Semester A.Y. 2025/2026 • SGAS/160/UN61/2025/FT




### 8. Committee Member: Adjunct Professor Program — FT UPNVJ

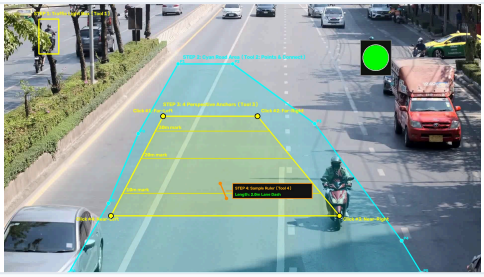
September 2025 • SERTF/546/UN61/2025/FT



# APPENDIX 2 — TECHNICAL ENGINEERING PROJECT PORTFOLIO

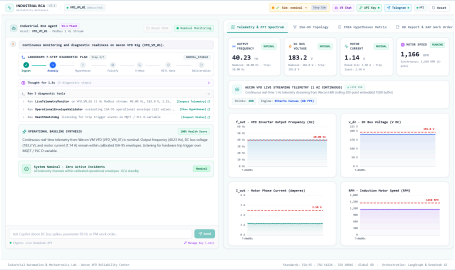
## 1. DeploySkripsi — ATSC Edge Runtime (NVIDIA Jetson & YOLOv11)

4-Point Perspective ROI Calibration, YOLOv11 & Sugeno ANFIS



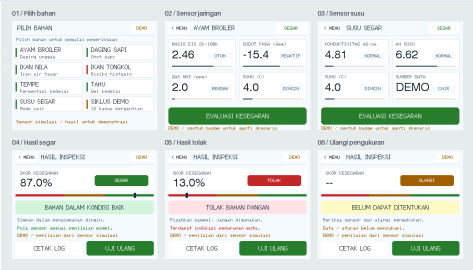
## 2. Industrial Root Cause Analysis (LangGraph & Modbus RTU)

Multi-Agent RCA Workflow Compliant with ISA-95 & ISO 14224



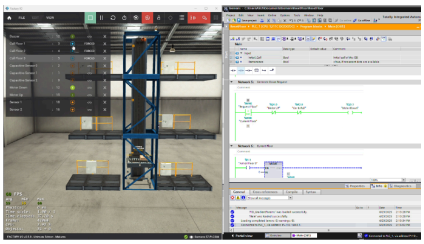
## 3. MBG Food Freshness & TFT Touchscreen GUI (ESP32-S3)

C99/Python Fuzzy Inference Engine on GUITION JC2432W328C



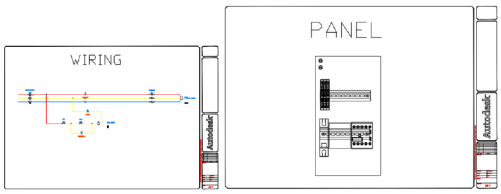

## 4. PLC-Based Multi-Floor Elevator (Siemens TIA Portal & Factory I/O)

3D Industrial Simulation & PLC Ladder Logic Control




## 5. Direct On Line (DOL) Motor Starter & Panel Layout

Power/Control Wiring Schematics & Panel Enclosure in AutoCAD Electrical




## 6. VisionLabs — Traffic Annotator & Multi-Model Label Fusion

GroundingDINO + YOLOv11 Fusion & Active Learning Priority Scoring

